

SmartMoo, An Agri-IoT Revolution



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Our daily diet includes dairy products in one form or the other, be it milk, ghee, butter or other milk based products. These play a very important role in maintaining the nutrient balance in our body. But, how do you know if the milk you drink is hygienic, with nutrients in proper portions, stored under correct conditions and is from a healthy cow? Definitely, a milk-packaging brand can answer these questions to some extent, but what about farmers, who have no idea about the milk's quality or contents? They would not even know if they are getting paid proportional to the quality of the milk or if their cow is under-nourished.

Milk is India's largest crop—India alone produces about 155 million tones of milk annually, and this number can increase further if the emerging markets perform well. So to resolve this issue, how about a full-stack Internet of Things (IoT) company that provides end-to-end dairy solutions? Here, an end-to-end solution covers tracking and monitoring of animal health, automated milking system equipped with smart machines, smart milk procurement peripherals, real-time payments and chilling milk using bulk milk coolers.

This is what a team of five co-founders with combined 90+ years of experience have formulated. They are called Stellapps, and their creative product is called SmartMoo.

Spelling out SmartMoo

The innovative applications of SmartMoo use the IoT, Big Data, Cloud, mobility and data analytics to improve agriculture supply chain parameters like milk production, milk procurement, cold chain, animal insurance and farmer payments. Primary focus is on data application and smart learning, using

SmartMoo by Stellapps is an end-to-end dairy technology. SmartMoo IoT router and in-premise IoT controller acquire data via sensors that are embedded in milking systems, animal wearables, milk chilling equipment and milk procurement peripherals, and transmit the same to Stellapps SmartMoo Big Data Cloud SDP, where SmartMoo suite of applications analyse and crunch the received data before disseminating it to various stakeholders over low-end and smart mobile devices.

smartSDP (service delivery platform)—the full-stack IoT solution to optimise dairy supply chain.

The IoT at the farm. Milk production-side IoT intervention includes sensors in the milking system. For example, an electronic milk meter monitors the functioning of milking machinery, animal wearables such as cow pedometers, radio frequency identification (RFID) ear tags and Android applications to scan and capture data from small farmers who cannot afford automation. This data from sensors is acquired by SmartMoo IoT router and in-premise IoT controller and transmitted to Stellapps SmartMoo Big Data Cloud SDP.

Over the Cloud. In Cloud servers, SmartMoo suite of applications analyses and crunches received data before disseminating the analytics and data science outcome to various stakeholders over low-end and smart mobile devices. Data acquired is used by Cloud-side analytics and machine learning algorithms for yield improvement, preventive healthcare, accurate oestrus detection, reducing inter-calving period, nutrition improvement, optimised animal insurance and reduction in the cost of operations. The data acquired at the milk production level by SmartFarms is used in the next hop of the supply chain.

At milk-collection centres. Milk procurement side includes sensors to analyse the milk quality, assess adulteration-limiting antibodies leaving the milk, assess the farmer's performance and save solid not



Fig. 1: Activity meter on a cow

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Smartmoo-Herdman	
	
HRB Code/Name	210841-Mahagauri09
Owner Code	104121058
Owner Name	Kamala
Age	5
Breed Name	Jersey
Status	Milking
Breeding Status	Open Breed
Open Period	15d
No of AI	1
Last Sire	9099992
Last Heat Date	2016-07-22 06:14:03
Expected Heat Date	18-08-2016
Expected Dry Date	14-03-2017

Fig. 2: SmartMoo milk production application—SmartMoo-Herdman